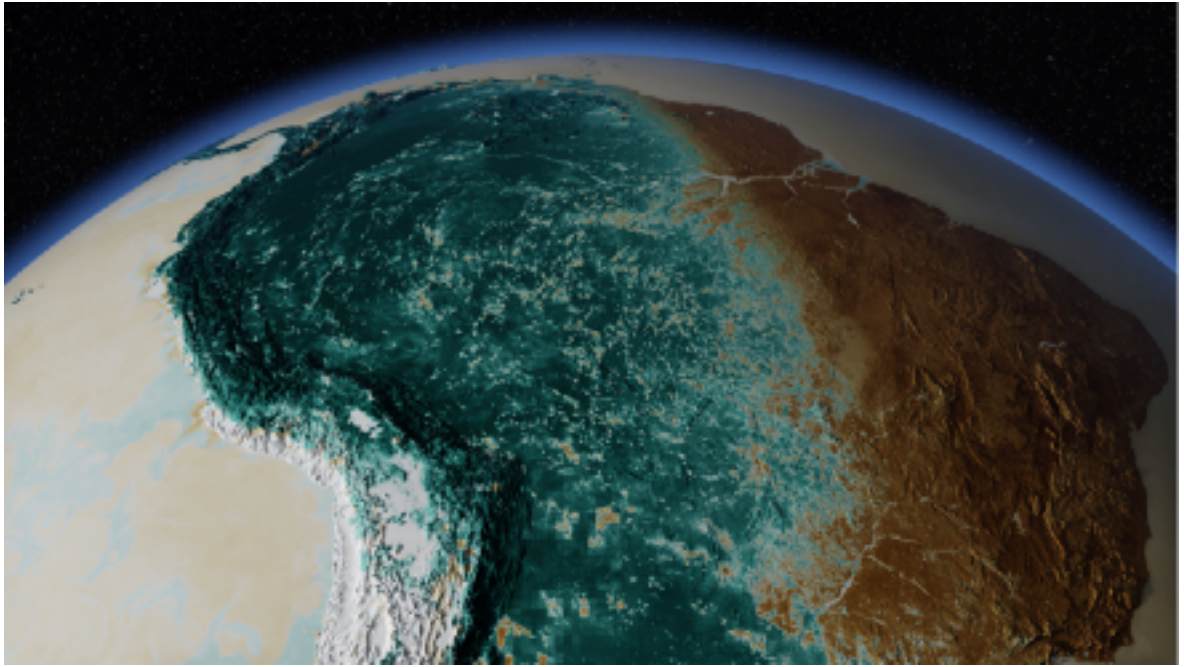
- It is impossible to measure everything everywhere all at once*

Appendix: we need models:

31

CO_2 flux over the Amazon basin from 1.25-km simulation



*now and for the next 40 years, every 10 seconds

- And, it is impossible to turn off clouds (for example) worldwide and see what happens…

Appendix: we need models:

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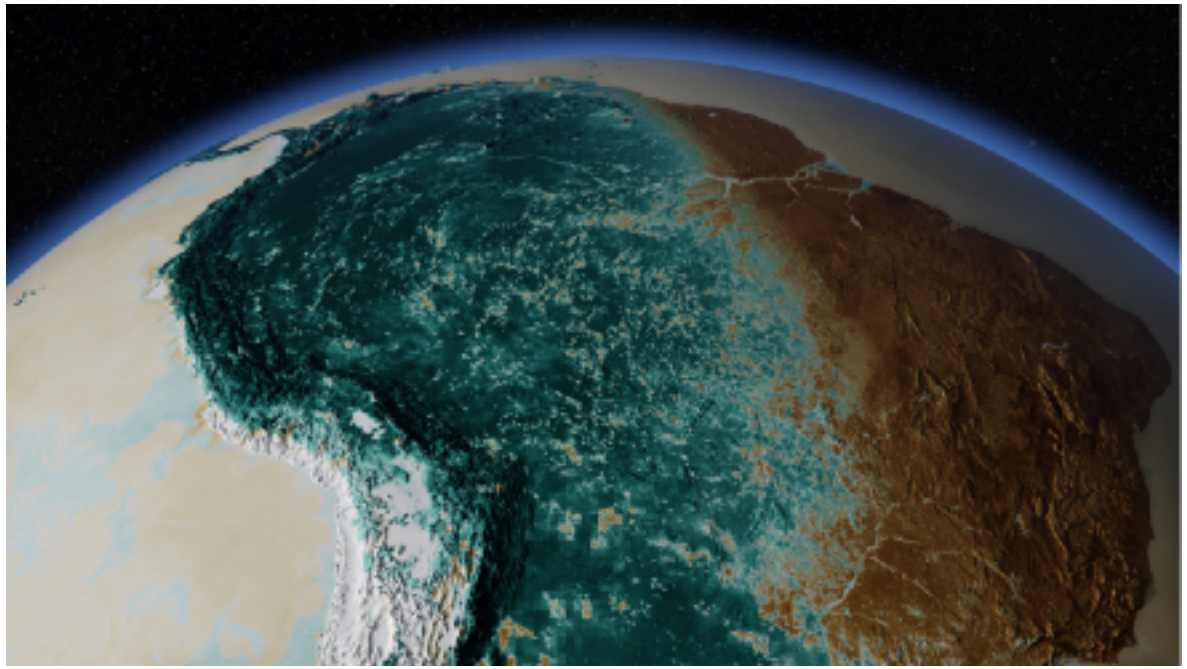
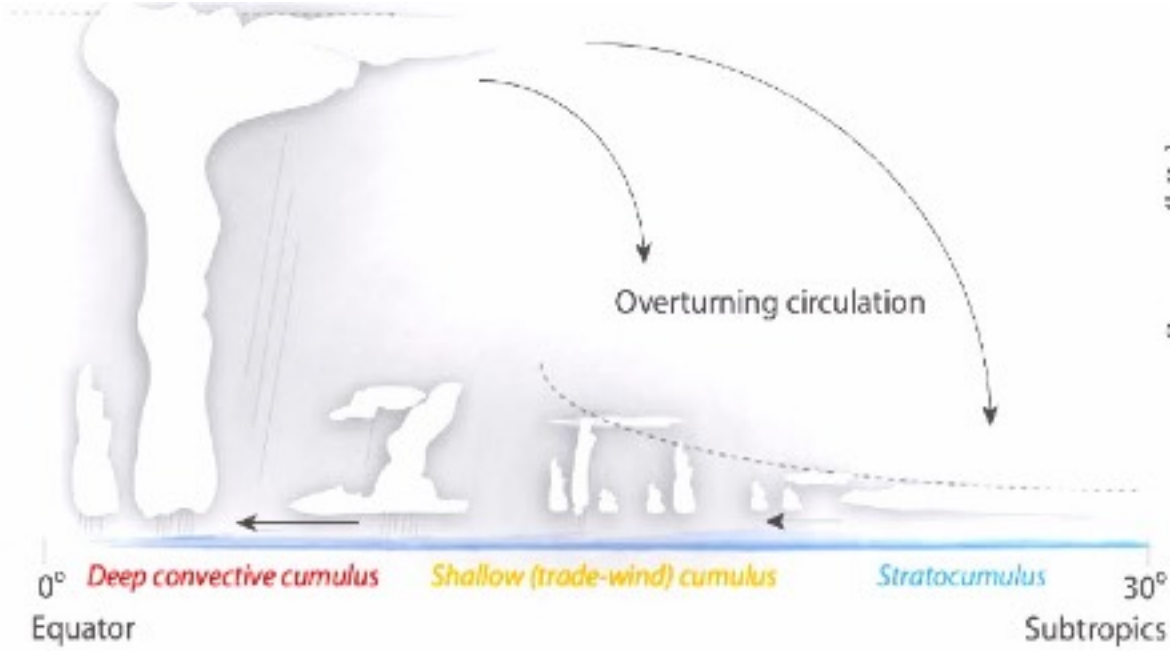


CO_2 flux over the Amazon basin from 1.25-km simulation

*now and for the next 40 years, every 10 seconds

We focus a lot on tropical clouds: why?

- Vis à vis radiation:
  - differentiating low and high cloud physics and feedbacks is important
- Importance of tropical clouds:
  - There are a lot of them!
  - Deep convection
  - Heavy precipitation
  - Large scale circulations



Nuijens & Jakob, *Clouds and Climate* (2020) (& Stevens & Hohenegger :))